Sohom Chatterjee

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Summary

CS undergraduate and ML engineer building production AI systems across document intelligence, NLP, computer vision, agentic workflows, and data analytics. Built OCR/document-extraction systems at Deloitte, behavioral scoring at HackQuest, and ML services for a live fintech platform at Lumeo. Focused on measurable outcomes and end-to-end ownership.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Rust, C, Solidity

ML/AI: PyTorch, TensorFlow, scikit-learn, Hugging Face, BERT, LangChain, LangGraph, LLMs, OCR, NLP, Computer Vision

ML Practice: Data Preprocessing, Feature Engineering, Fine-Tuning (LoRA/QLoRA/PEFT), Evaluation, SHAP, LIME, Grad-CAM++, MLOps, K-Means, Forecasting, RFM

Backend/Data: NestJS, Node.js, FastAPI, Flask, REST APIs, Prisma, PostgreSQL, MongoDB, Redis, BullMQ, Docker, CI/CD

Cloud/Web: AWS, GCP, Vercel, Linux, Git, React, Next.js, Tailwind CSS

Experience

Lumeo (Oxify Labs Tech Pvt. Ltd.) Kolkata, India
Co-founder & Chief Executive Officer *Feb. 2026 – Present*

- Own product/backend architecture: **150+ REST endpoints, 79 Prisma models** on PostgreSQL, Redis/BullMQ queues, double-entry ledger, and Python/FastAPI ML services for document parsing, compliance, and reconciliation.
- Rewrote latency-critical NestJS services in **Rust**, cutting container image size by **95%** and lowering p95 latency on hot paths; lead a **4-engineer** team building the AI-native financial OS.
- Grew the product to **40,000+ waitlist signups** with zero ad spend and secured enterprise/design-partner traction pre-launch.

HackQuest Remote
Developer Advocate (Internship) *Nov. 2025 – Present*

- Built an **ML pipeline** scoring learning progress from in-platform user actions, improving point-attribution accuracy by **50%**; mentored **25+ students** on Web3/Solidity and wrote technical tutorials.

Deloitte Remote (APAC)
Machine Learning Engineer (Freelance) | *Client: Integra LifeSciences Holdings Corporation* *Jun. 2025 – Sep. 2025*

- Built an OCR/document-intelligence workflow to classify and extract information from messy healthcare/business documents; added layout detection, denoising, deskewing, and post-OCR processing, cutting processing time by **20%**.
- Delivered **4x ROI** through data-driven campaigns across legal, finance, and marketing; unified datasets with automated data-cleaning pipelines.
- Reduced healthcare-retail fraud by **17%** using SQL, K-Means, and predictive models; increased CRM growth by **20%** through RFM segmentation, demand forecasting, and market-basket analysis.
- Improved client retention using CSI/NPS metrics built with **20+ advanced SQL analytics**; built centralized dashboards.

Projects

Project Apex DB | *Rust*
Open-source embedded database for AI agent memory, written in Rust. Logs every change to a write-ahead log, isolates each tenant's records, and applies retried writes exactly once. project-apex-db.vercel.app

AETHER AI | *FastAPI, PyTorch, Hugging Face, PEFT/LoRA, Next.js, Celery, PostgreSQL, Redis, Docker*
End-to-end multimodal training platform: dataset connectors, schema/quality/lineage inspection, LoRA/QLoRA/adapter/prompt tuning, 0.85 F1 baselines, versioned model registry.

Explainable AI for Plant Disease Detection | *CNN, SHAP, LIME, Grad-CAM++, TensorFlow*
Co-authoring XAI-enhanced CNN research; 20-paper literature survey and an interpretable visual-explanation pipeline for plant disease classification.

Serene Mind / Zelda | *Python, PyTorch, OpenCV, AWS, FaceNet, MTCNN*
Serene Mind: multimodal text/voice/facial emotion detection with fused risk scoring. Zelda: facial authentication with anti-spoofing.

Archi.dev / Shortlist AI / AutoFi / DACAP | *LangChain, NLP, Celo, Solidity, Rust*
LLM + graph-reasoning microservice architect; NLP resume ranking; multi-agent treasury/risk automation; AI smart-contract auditor.

Open Source Contributions

- Fixed a TypeScript import-type normalizer performance bug in **Graphify** (sorted-index lookup); merged in v0.9.56, issue #3359.
- Fixed three JSON-formatting failures in **Inspect AI** grading templates; filed issue #5276, fix built on PR #5256.

Education & Achievements

Sister Nivedita University Kolkata, India
B.Tech in Computer Science & Engineering (Data Science) *Aug. 2024 – 2028*

SIH 2024: National Finalist | **Google Solution Challenge:** Awardee, 2025 | **Gemini Campus Ambassador:** 2025
Certifications: MLOps for Generative AI; Transformer Models & BERT (Google Cloud); Full Stack & AI/ML Developer (Sheryians)
Community: GeeksforGeeks SNU Technical Executive (mentored 15+ teams); LNC Network OSS